Created in Cambodia, November 2012 (Revised November 2014)

Rice Ragged Stunt Virus

Recognize the problem

Rice plants infected with Rice ragged stunt virus (RRSV) typically show stunting, yellowing leaves with rough edges or twisted tips, and vein swelling. Galls are caused by swollen phloem cells and develop on the leaf blades and sheaths. Leaf edges are uneven and the twisting gives the leaves a ragged appearance.

Ragged Virus on leaves. (Photo by IRRI (CC BY-NC-SA))



Background

Ragged stunt is caused by a virus and is transmitted by the Brown planthopper (BPH) nymphs and adult. BPH adults move from ragged stunt-infected fields to newly planted rice fields, and so spread the virus. Other sources of disease inoculum consist of infected host plants, crop stubble and volunteer rice plants. RRSV infects many graminaceous plants in inoculation tests using BPH. Natural infection of weeds and cereals other than rice, however, is rare, because BPH feeds only on rice.

Ragged virus on leaves. (Photo by IRRI (CC BY-NC-SA))



Management

Once infected by the virus, a plant cannot be cured. Therefore, prevention is the most effective management option.

- Plant rice cultivars with resistance to BPH, if available
- Synchronize the rice planting and rotate with secondary crops. This requires community-wide involvement.
- Remove and burn or plough-under infected stubble
- Encourage natural enemies e.g. water striders, mirid bugs, spiders, and various egg parasitoids. If the population of these exceeds that of BPH, do not apply insecticides.
- If not, use an insecticide to control BPH which transmits the virus, e.g. abamectin (0.5mL/L water), Fenobucarb (20-30mL/8L water) or permethrin (1mL/L water)

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > ***Rice Ragged Stunt Virus***

The recommendations in this factsheet are relevant to: Cambodia

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